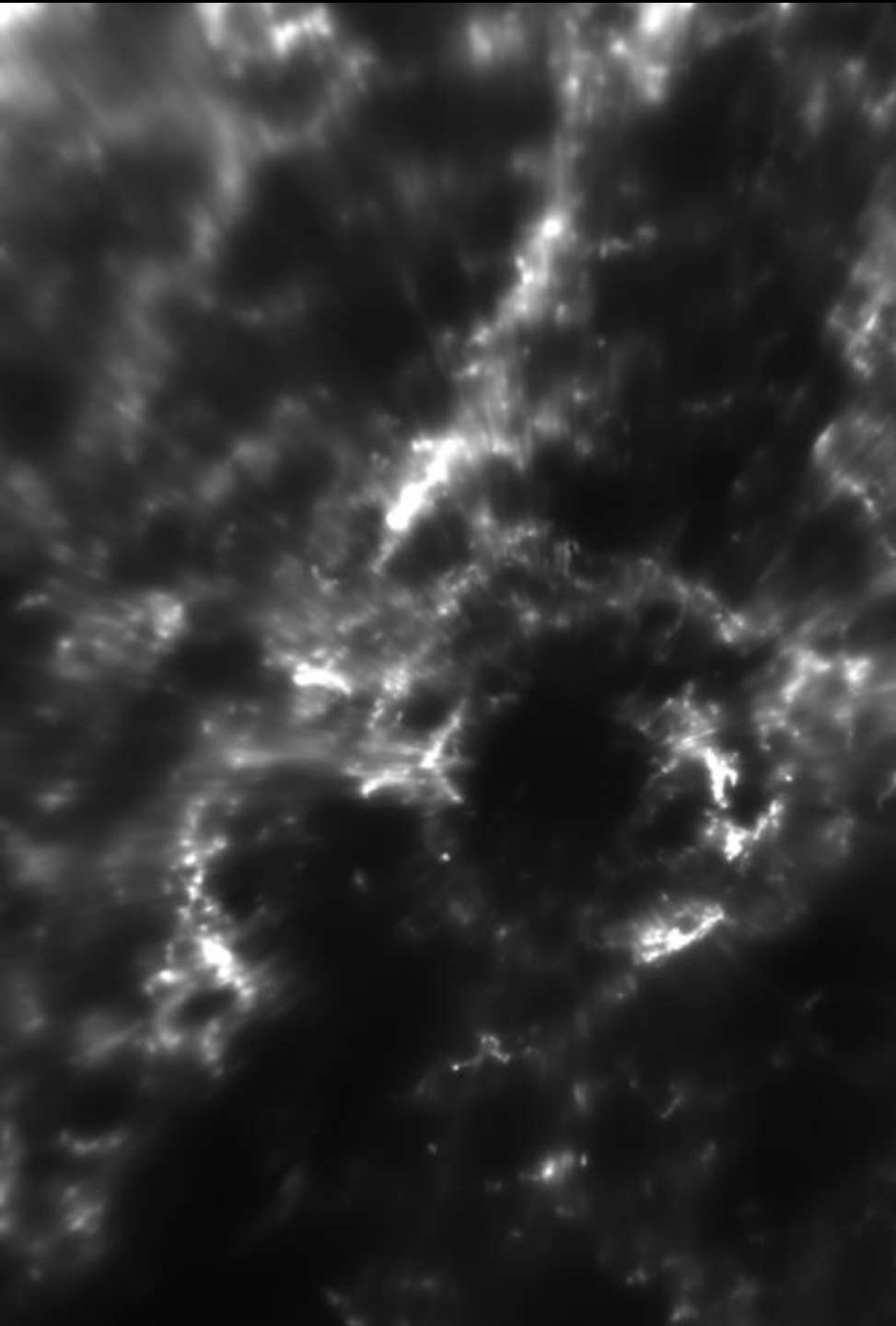




The Superclouds of the Local Milky Way

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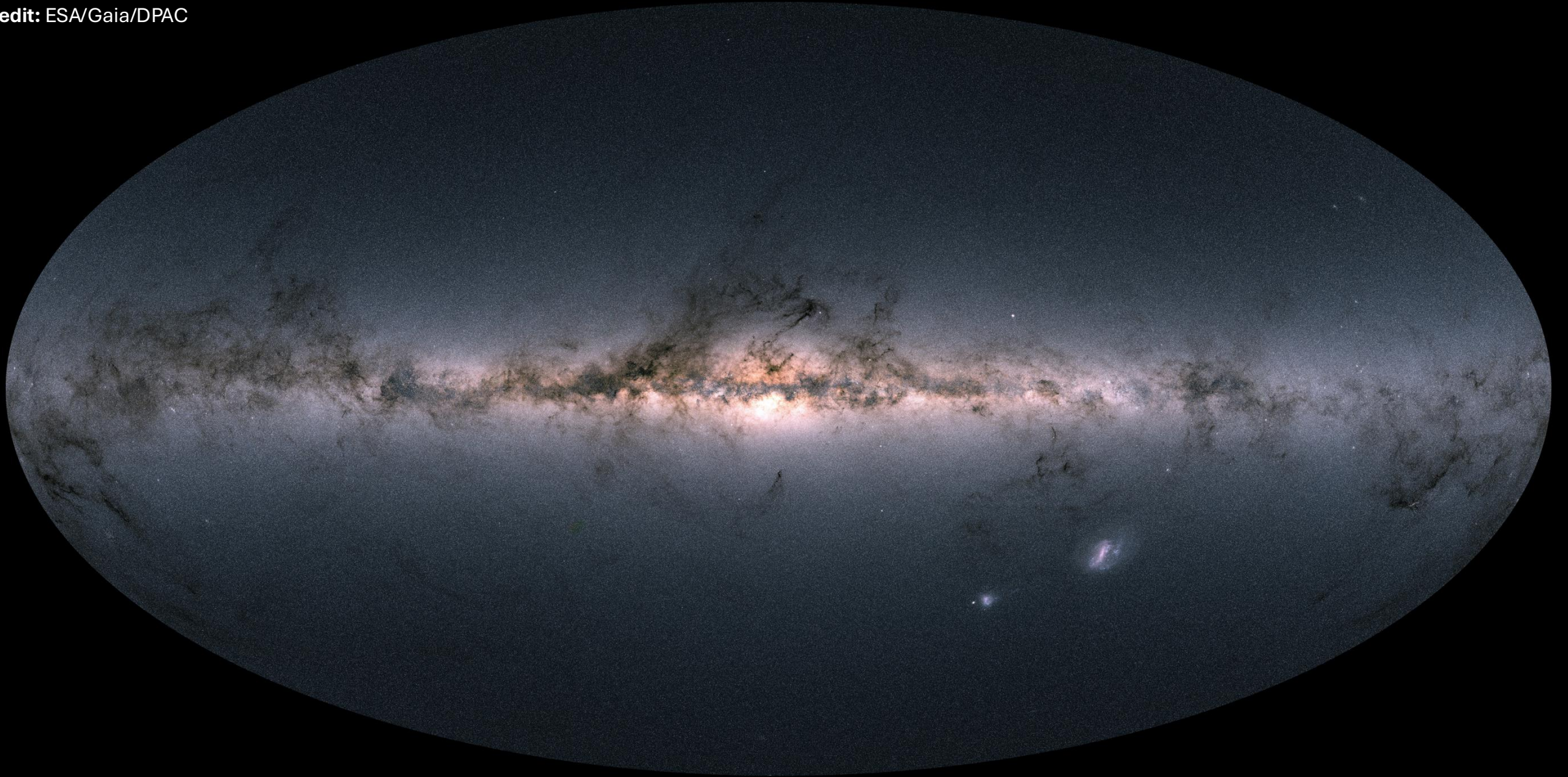


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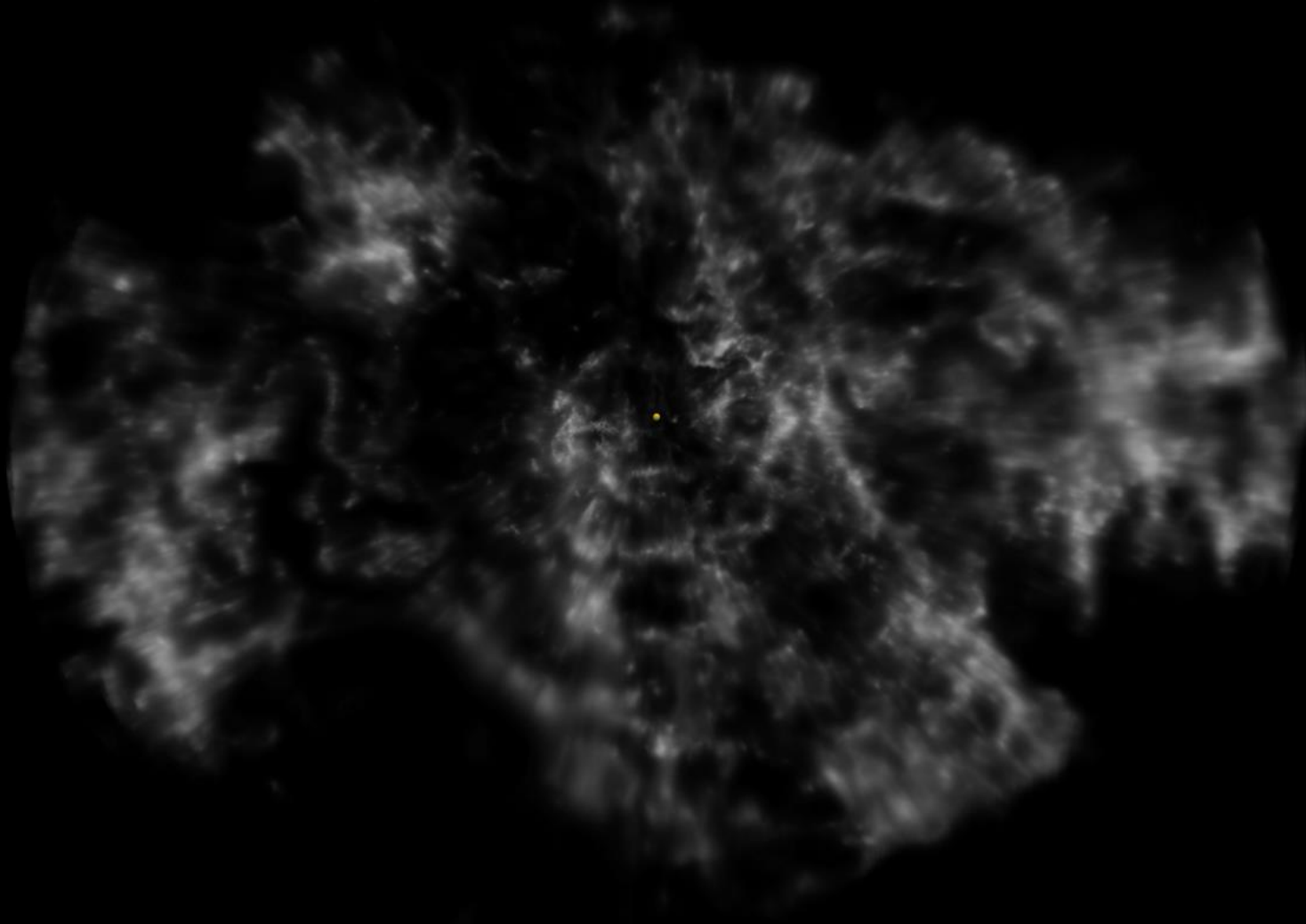


Gaia's All-Sky View

Credit: ESA/Gaia/DPAC

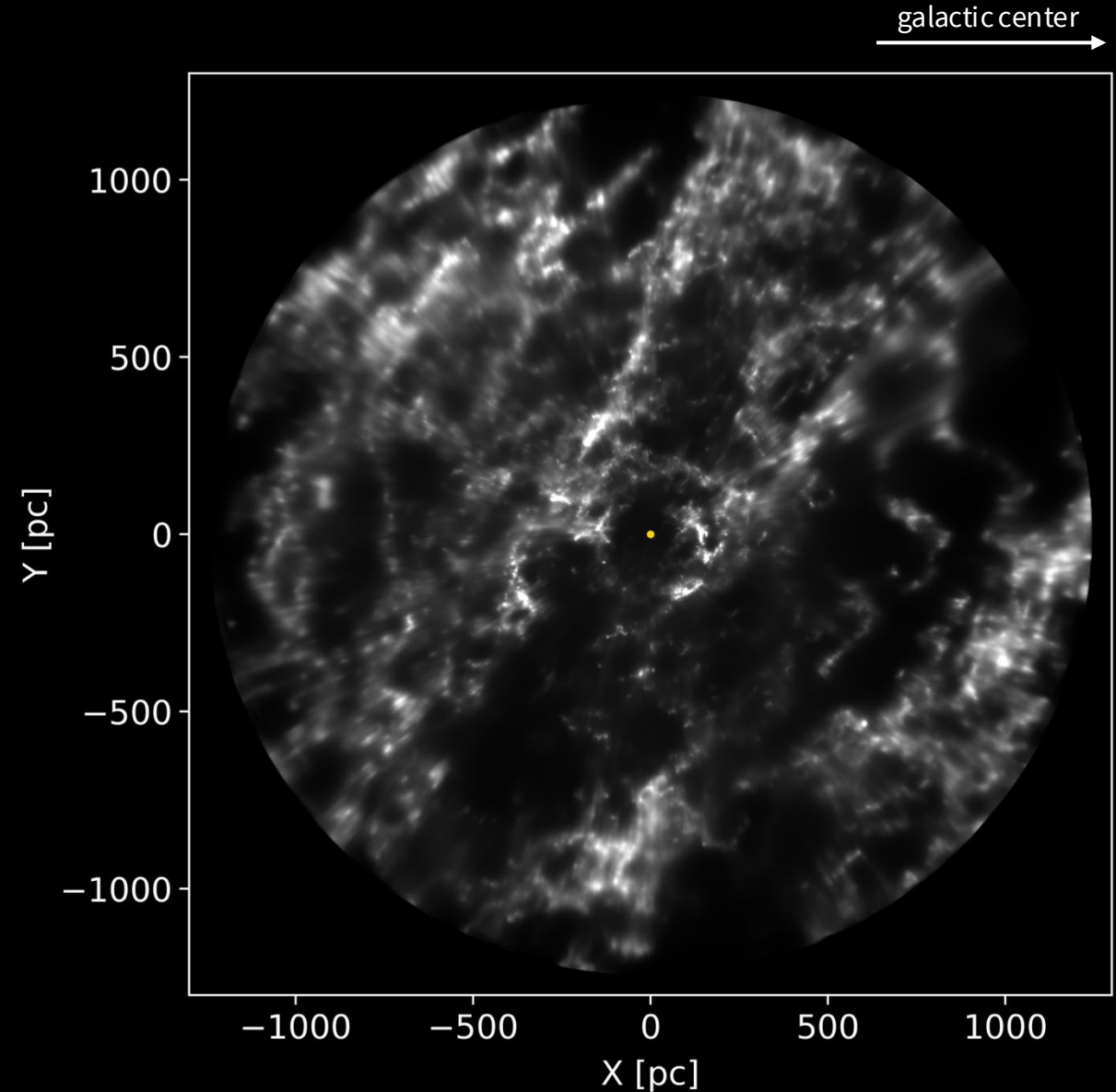


3D Dust Distribution



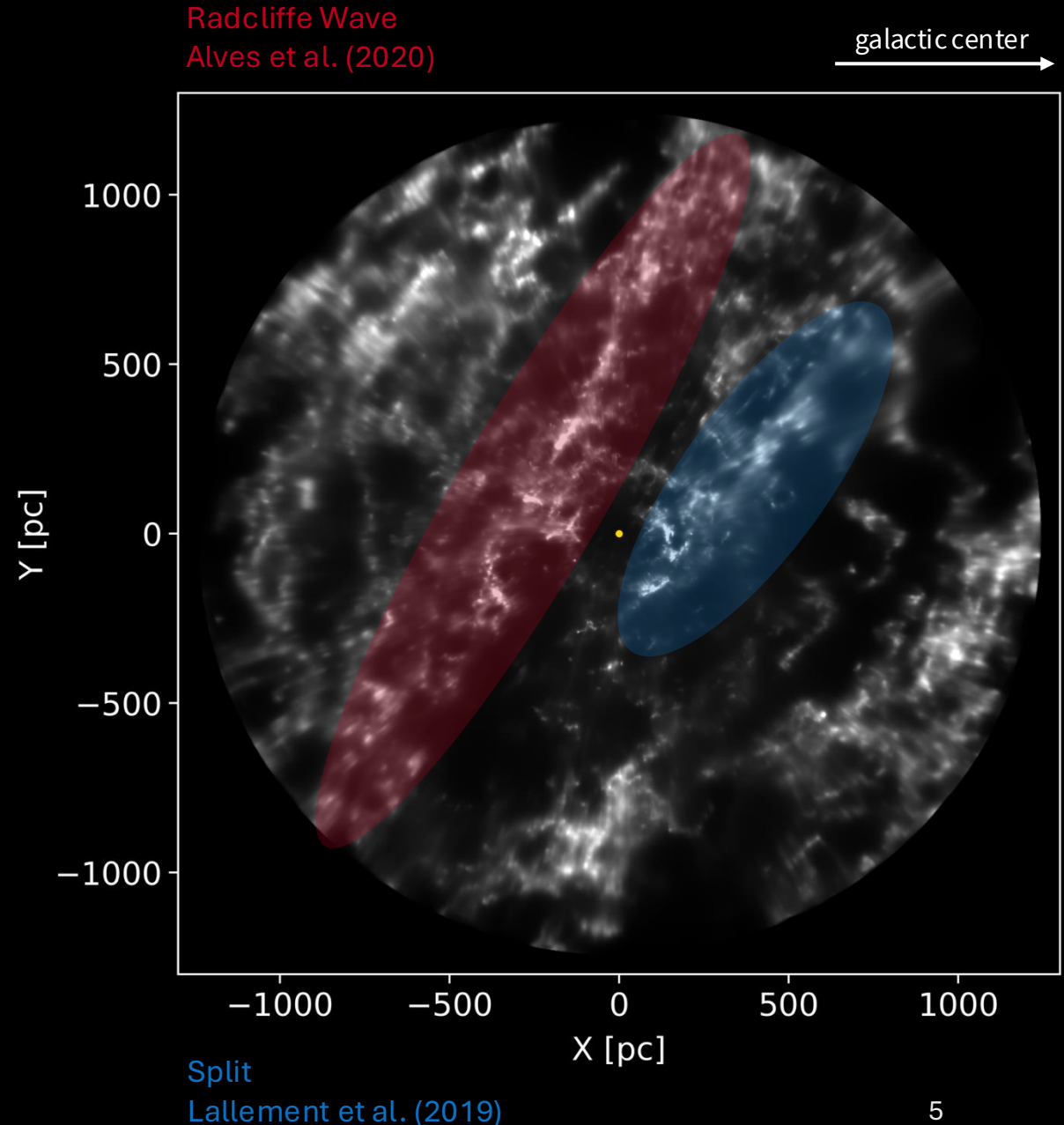
Local MW in 3D

- 3D dust map by Edenhofer+2024
- stellar distance and extinction estimates derived from Gaia spectra
- spatial distribution of dust extinction out to 1.25 kpc



Local MW in 3D

- 3D dust map by Edenhofer+2024
- stellar distance and extinction estimates derived from Gaia spectra
- spatial distribution of dust extinction out to 1.25 kpc
- **motivation:** if possible, classify (visible) large scale structures



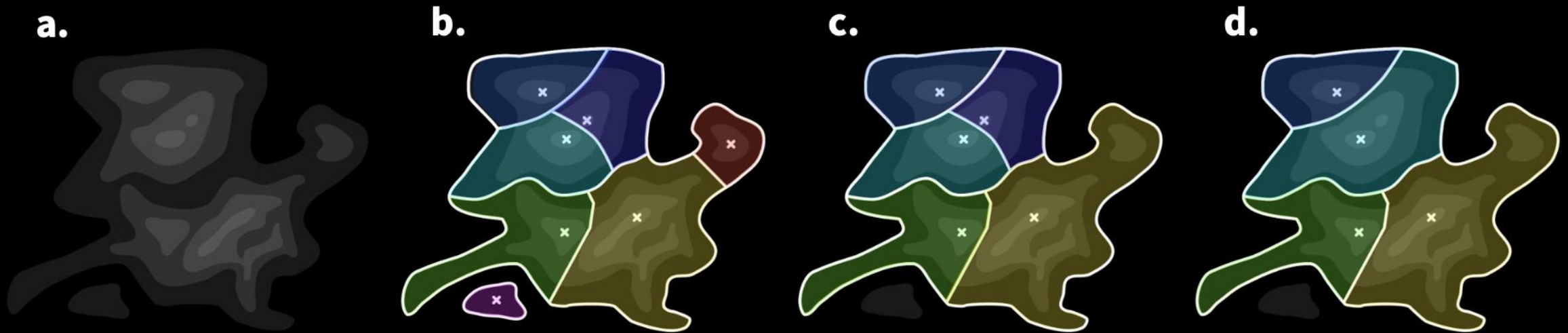
Early History of Superclouds

- some of the largest recognized structures within the interstellar medium
- first mentioned by Elmegreen & Elmegreen (1983)
 - neutral atomic hydrogen superclouds in spiral arms in external galaxies
 - masses between 10^6 and $10^7 M_{\odot}$ and kiloparsec lengths
- further work done by Elmegreen & Elmegreen (1987)
 - identified neutral atomic hydrogen superclouds in the inner Galaxy
 - noted spatial correlation with giant molecular cloud complexes

**How can we attempt to classify
such structures?**

Structure Finding Algorithm HOP

- originally used to identify groups in N-body simulations
- adapted by Colman et al. (2024) to find molecular clouds in simulations
- analyzes voxels around peak patches and border densities for segmentation

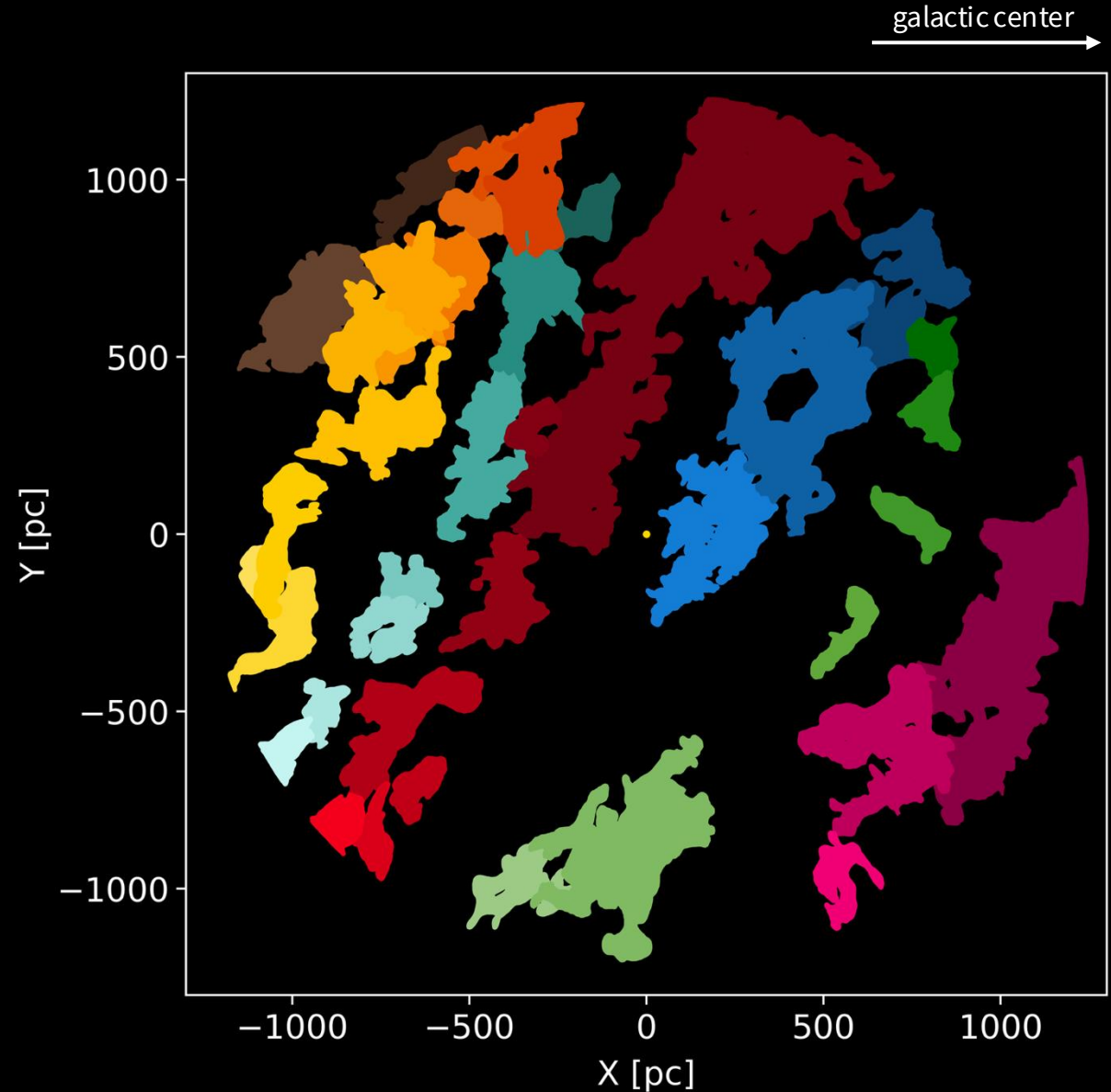


3D HOP Cloud View



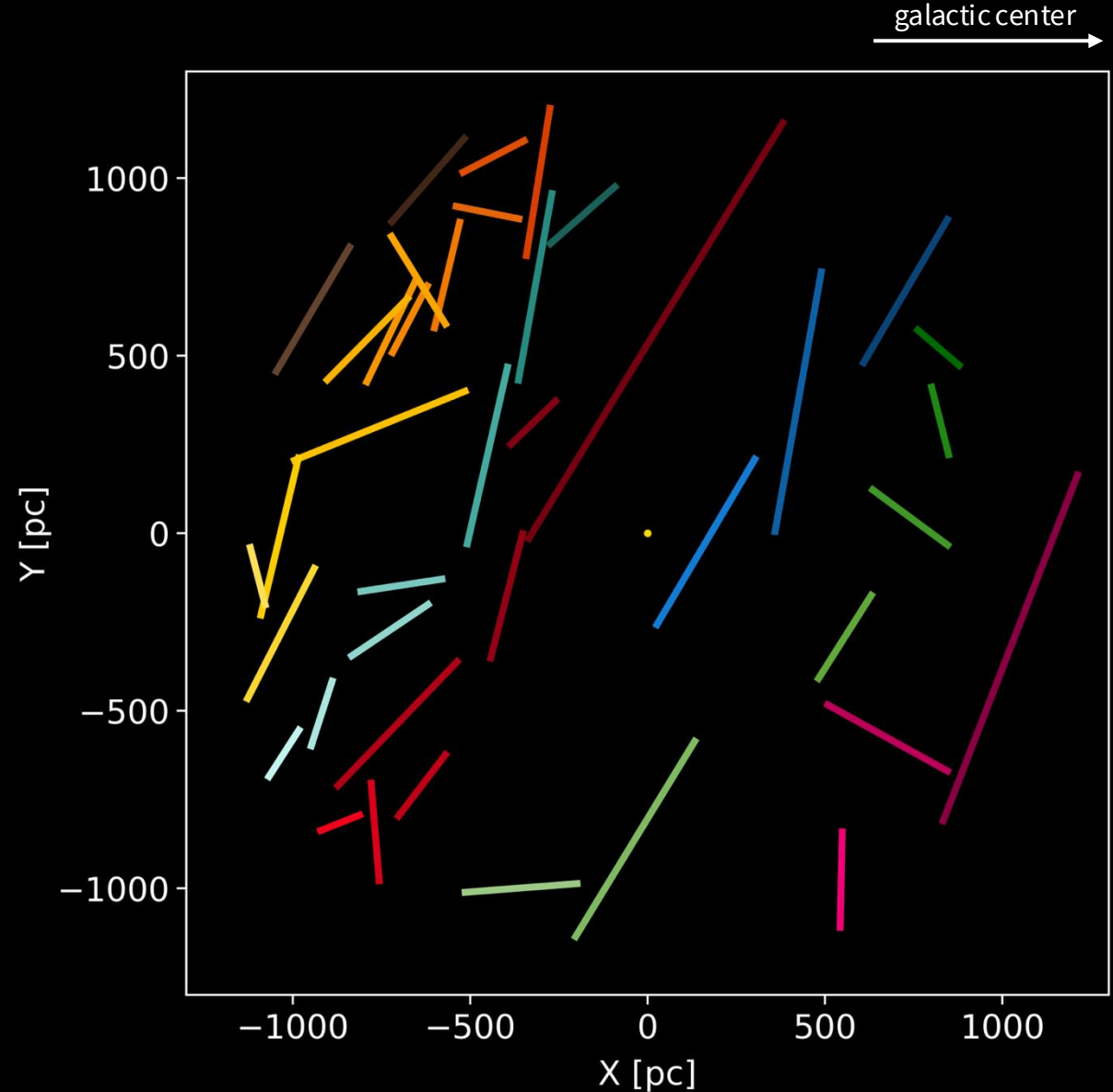
Segmentation Results

- 40 clouds identified by HOP in the dust map
- masses range from 10^4 to $10^6 M_{\odot}$
- show quasi-parallel alignment
- most prominent in **turquoise**, **red**, and **blue** cloud groups



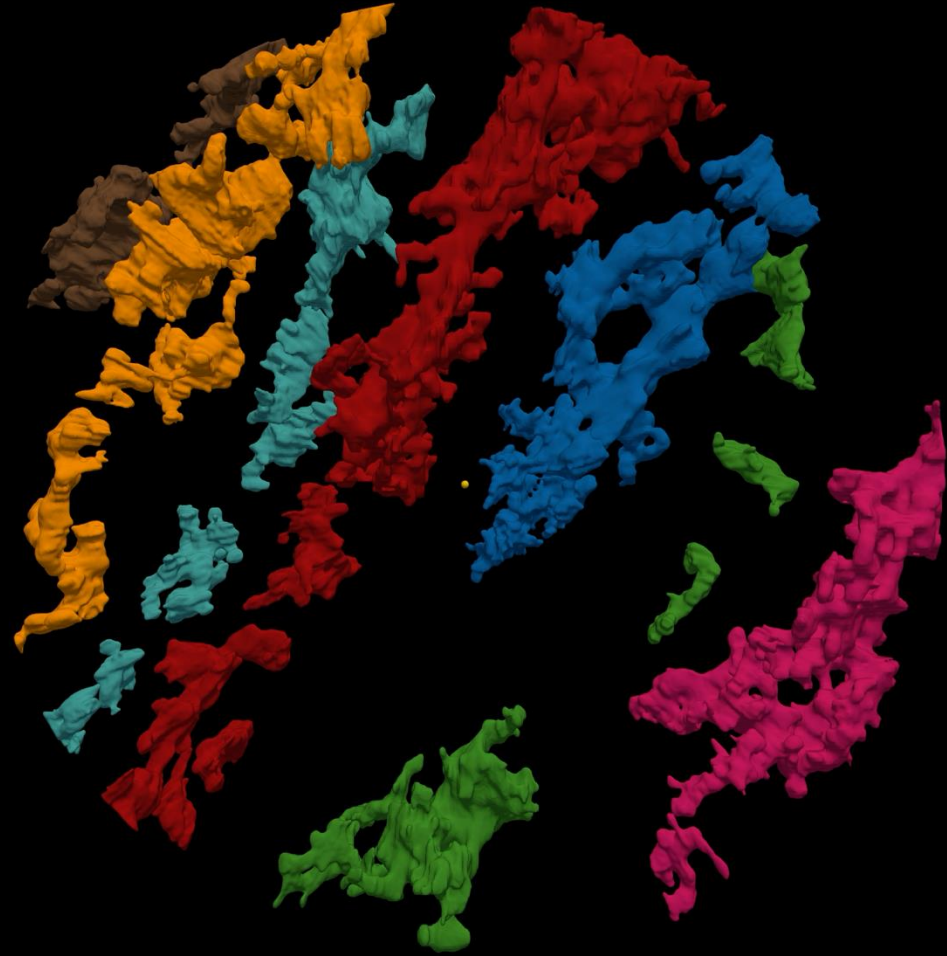
Average Orientation

- face-on orientation calculated via principal component analysis
- average cloud orientation face-on of $22.6^\circ \pm 25.5^\circ$
- larger clouds show orientations closer to 30°
- smaller clouds scatter more around average



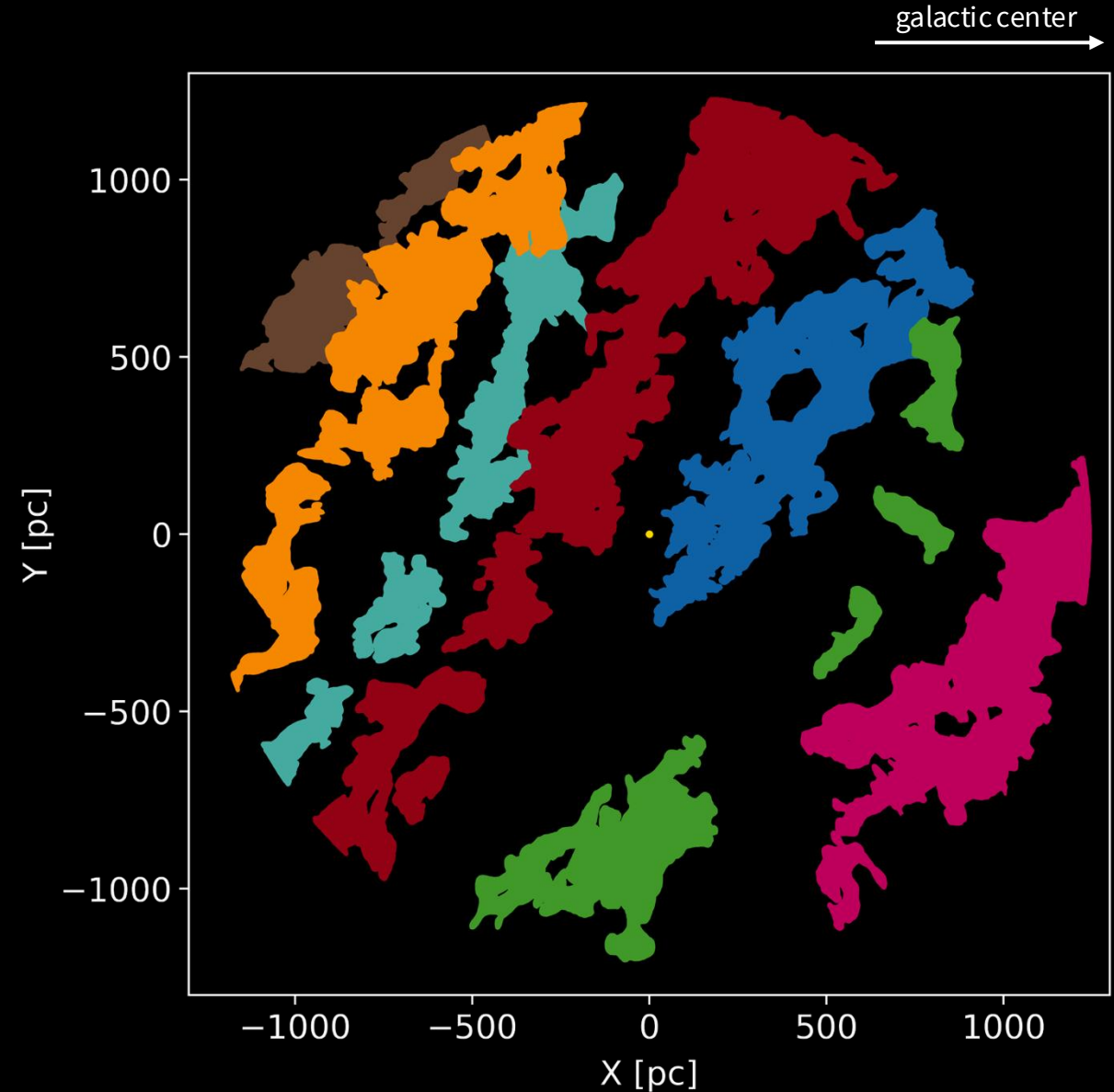
Connecting Clouds to Superclouds

3D Supercloud View



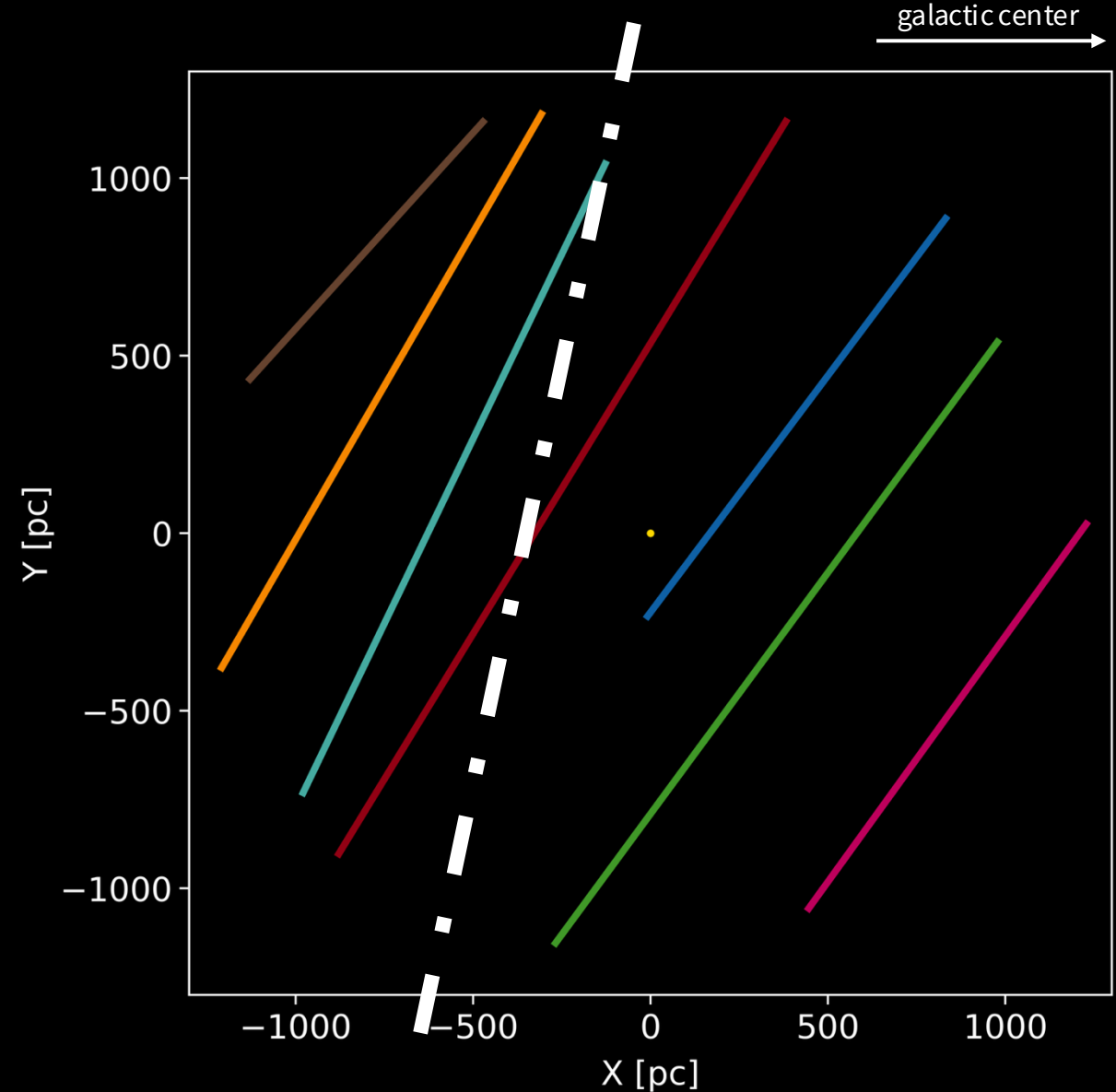
Connection Results

- 40 clouds are connected to 7 superclouds
- two superclouds previously known
 - Radcliffe Wave (**red**)
 - Split (**blue**)
- 5 additional clouds defined



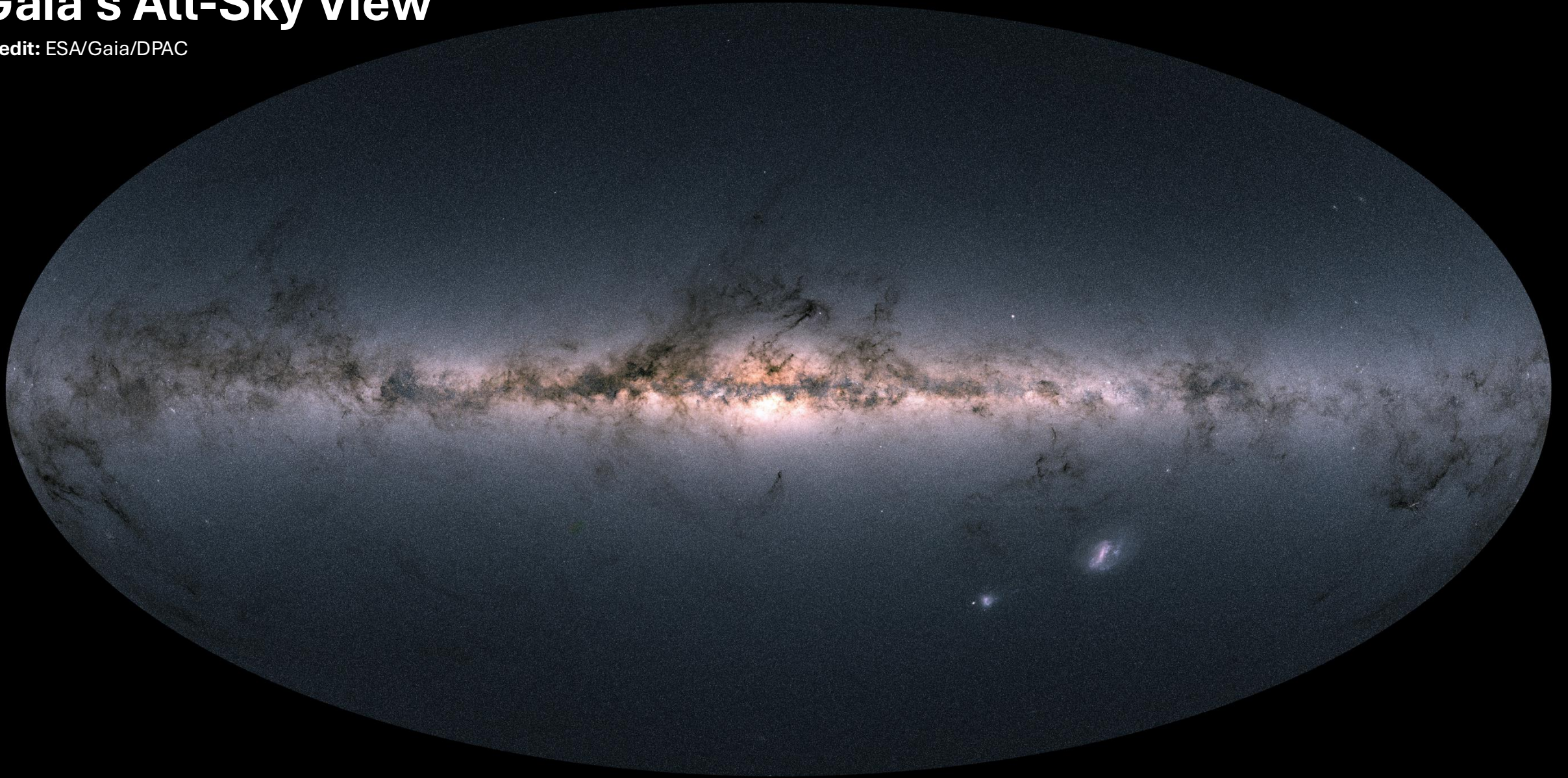
Average Orientation

- orientation computed via PCA of all cloud components
- average supercloud orientation close to 30°
- average spiral arm angle $\sim 10^\circ$
- unlikely to be associated with spiral arms



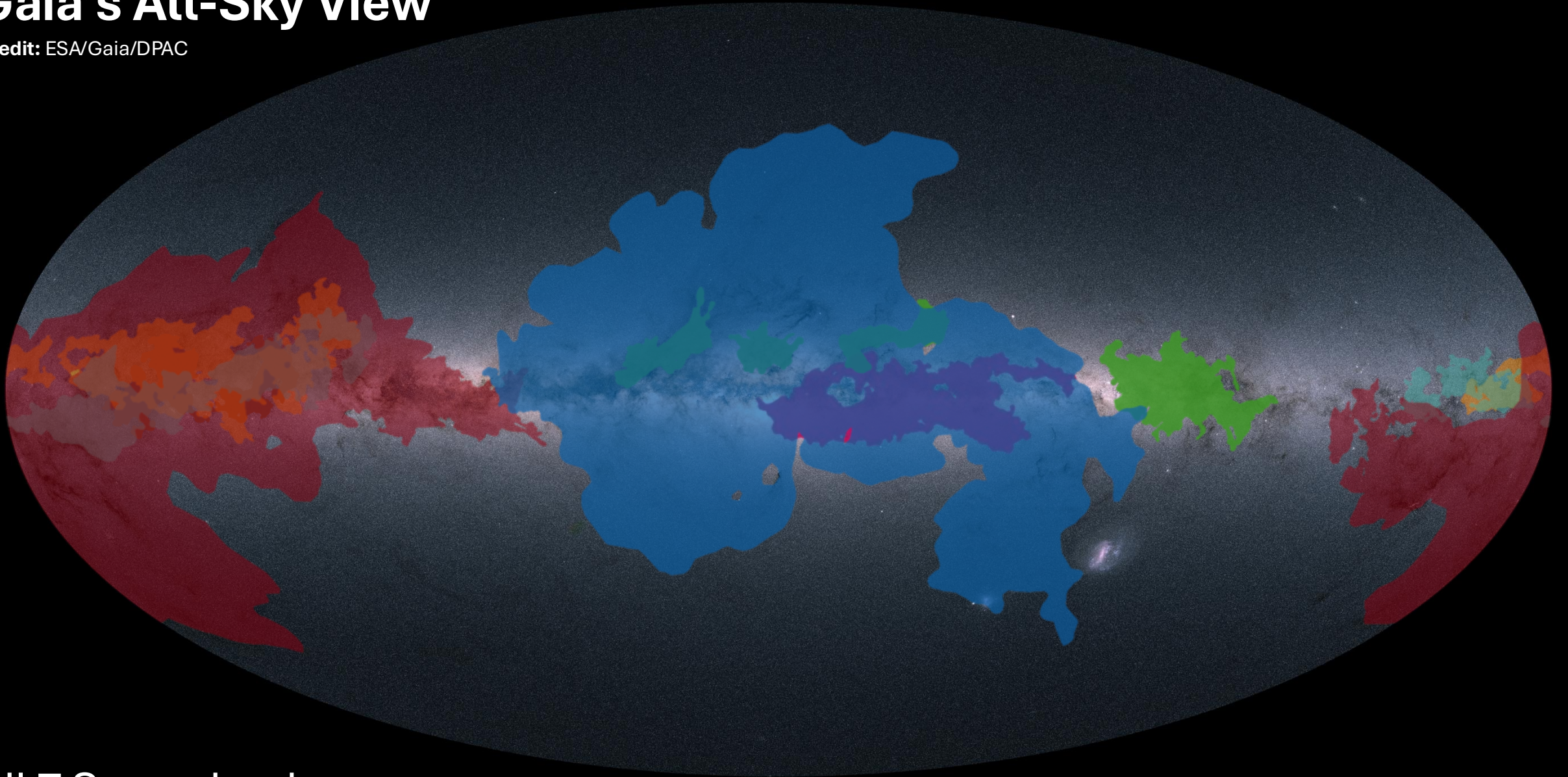
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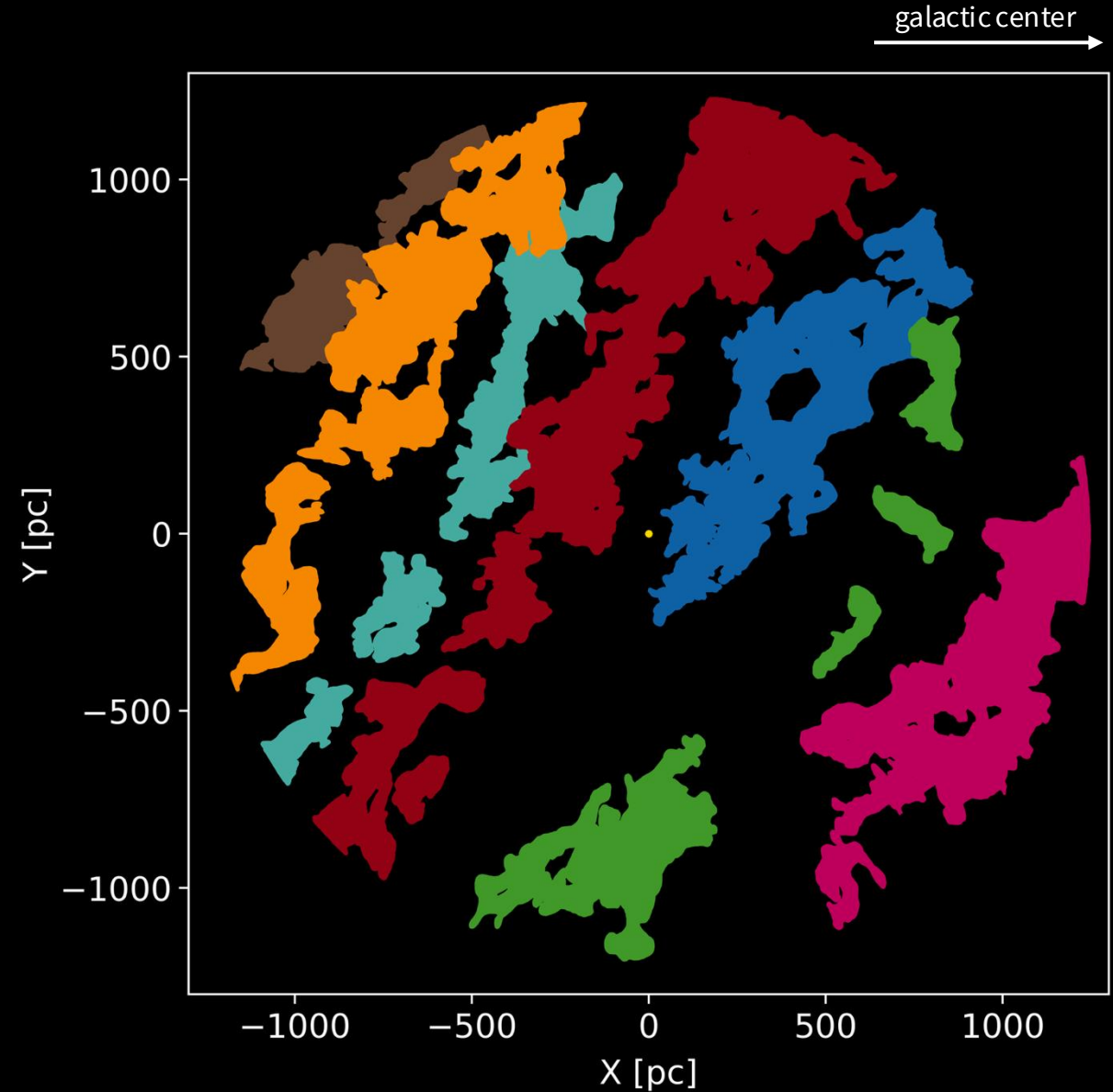


All 7 Superclouds

Kormann et al. (2026), Alves et al. (2020), Lallement et al. (2019)

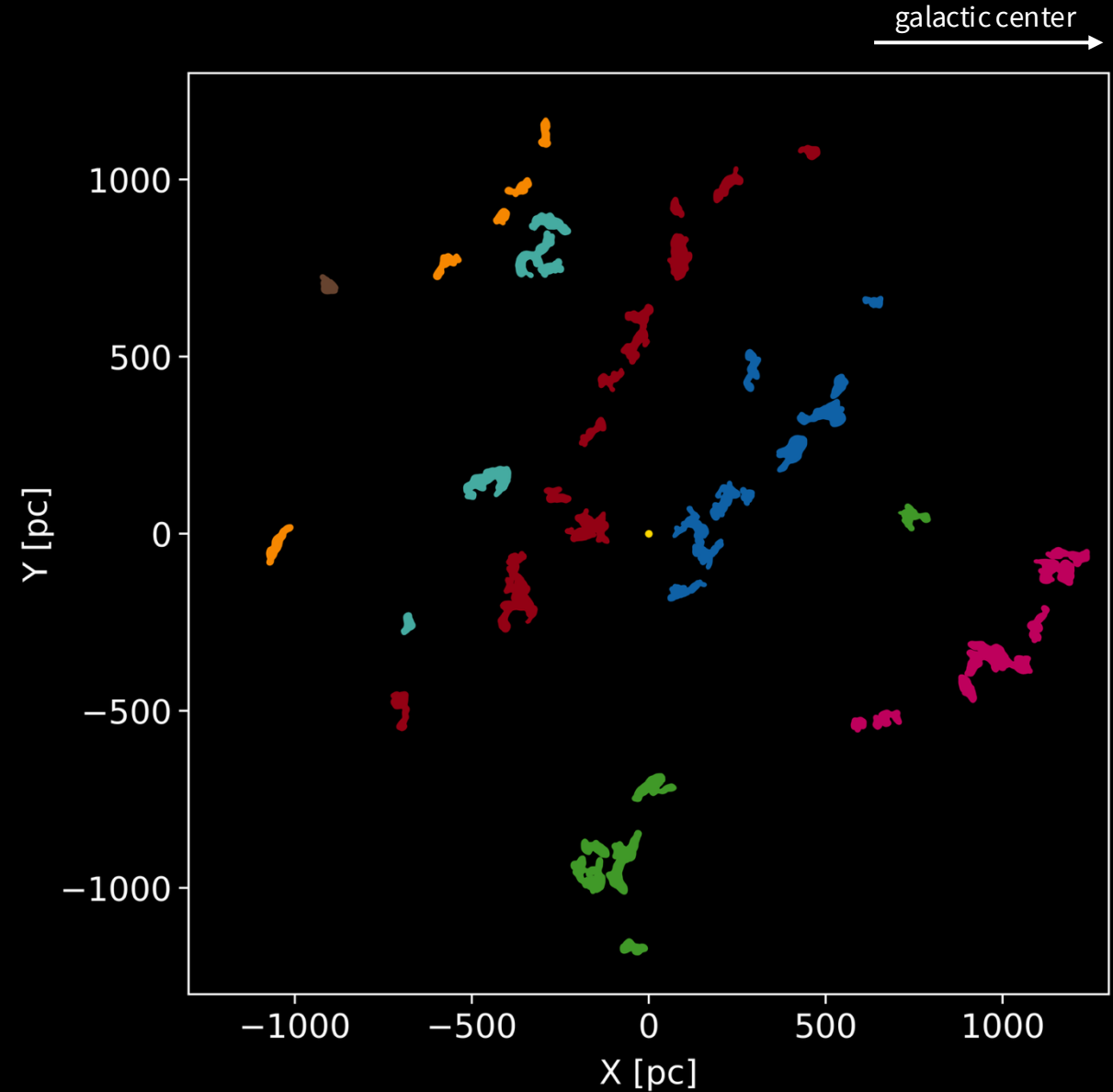
Current Work

- analyzing the internal density structure of the superclouds



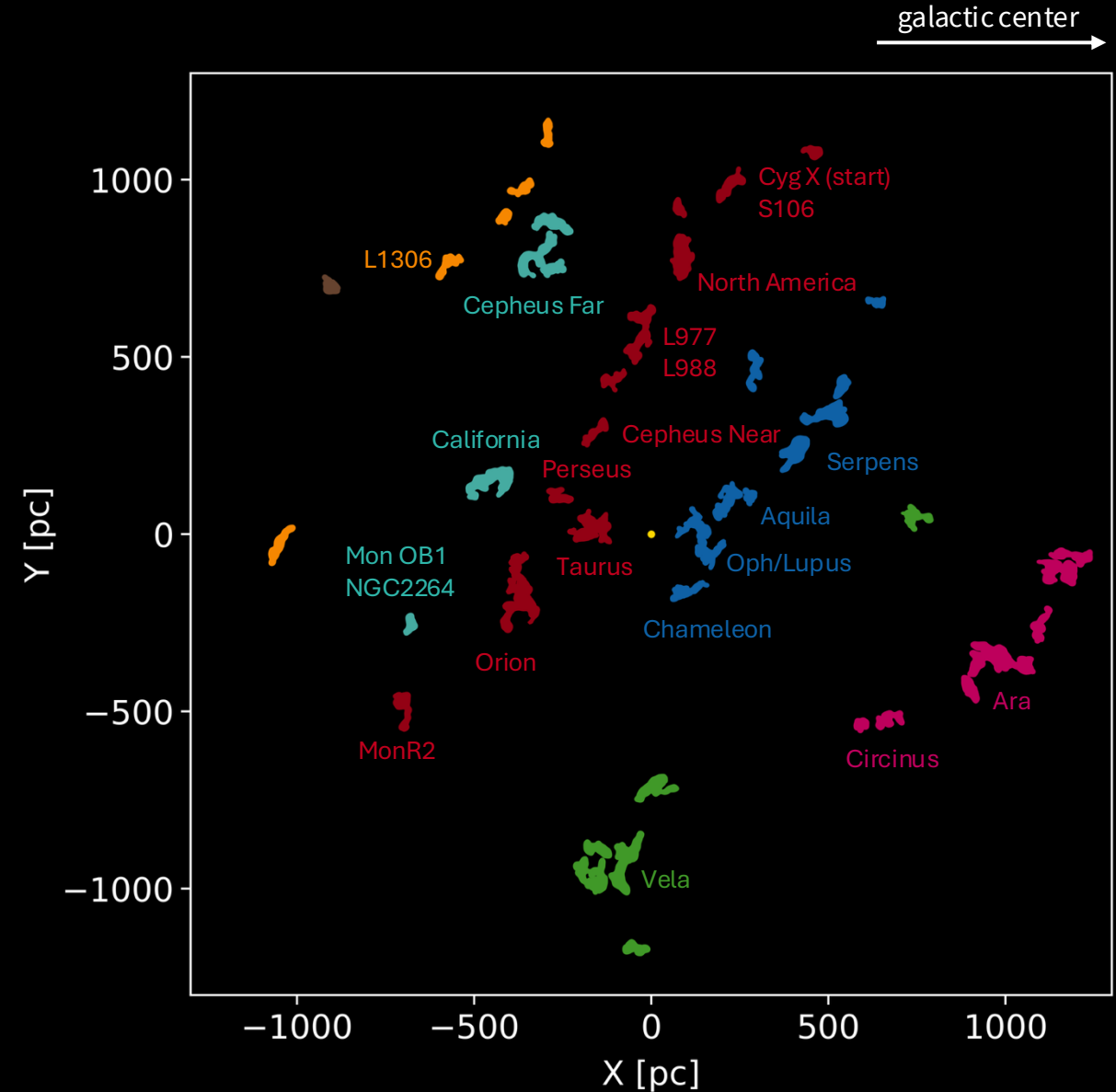
Current Work

- analyzing the internal density structure of the superclouds



Current Work

- analyzing the internal density structure of the superclouds
- correlation with known star-forming regions in the solar neighborhood
- the “superclumps” seem to show similar distance distribution per supercloud



Conclusions

- identification of 7 superclouds in the local Milky Way, with five previously unknown structures
- new insights into the distribution and properties of the most massive structures in interstellar medium
- a step towards connecting local and galactic structures
- superclouds likely play an indirect but important role in the process of star formation

Kormann et al. 2026

